Shayan Sadigh

Location: California, United States
Email: ssadigh.contact@gmail.com
Github: https://github.com/shayanpersonal (150+ stars)

Summary

- Former Amazon Deep Learning Engineer. Worked on SageMaker Neo team. Returning from career break and open to role switch.
- Won 10th in $1.5 million Kaggle competition - designed custom MVCNN-RNN neural net (Github link)
- Proposed new deep learning layer - "checkered subsampling" (arXiv paper)
- Ranked 1st globally (tied) in competitive arena shooter Splitgate (username "Moejoe").

Work Experience

Santa Barbara City College
Adjunct Professor of Computer Science (Aug. 2023 - Dec. 2023)

- Walked students through various aspects of designing and training deep learning models. Discussed linear layers, nonlinearities, normalization, convolution, aspects of transformers such as attention layers and embeddings. Discussed backpropagation and loss functions.
- Implemented k-nearest-neighbors and small neural networks in numpy and PyTorch for homework. Trained models on MNIST and tiny-shakespeare.
- Taught CS114: Intermediate Python. SBCC is frequently ranked among the top five city colleges in the United States.

moebot.ai
Creator (Jan. 2023 - Sep. 2023)

- Built GPT-4 based AI agent for Discord servers. Makes discrete decisions such as whether to respond or send emoji reactions via prompt engineering. Stores and recalls rudimentary "memory" of server history.
- Backed by Redis. Backend provisioned in Google Cloud. Generates revenue through Stripe transactions.
- Project put on hiatus after Discord announced an official LLM chatbot "Clyde."

Amazon Web Services (AWS)
Deep Learning Engineer (L5) (Mar. 2020 - Dec. 2020)

- Helped integrate Apache TVM deep learning compiler into Amazon SageMaker's PyTorch fork.
- Resolved open issues in the team's PyTorch and XGBoost forks.

Palo Alto Networks
Senior Staff Software Engineer (Oct. 2019 - Mar. 2020)
Staff Software Engineer (Sep. 2018 - Sep. 2019)

- Awards: Project of the Quarter - Machine Learning Antivirus
- Proposed and implemented a new stack-based pattern matching algorithm to improve threat coverage in PAN-OS, written in C. Increased our security effectiveness rating on NSS Labs firewall tests.
- Helped implement machine learning models such as XGBoost on resource-constrained firewall in C.
- Solved dozens of open tickets, including feature requests from data science and product managers.
- Brought up and helped train 2 junior engineers.

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Other
Spring 2018 : Teaching Assistant @ UCSB (CS176A - Intro to Computer Networks)
Summer 2017 : Software Engineer Internship @ Palo Alto Networks
Summer 2016 : Software Engineer Internship @ Thermo Fisher Scientific
2015 : Web developer @ UCSB Brain Initiative

Projects

- stacked-autoencoder-pytorch (100+ stars) - a popular PyTorch autoencoder.
- Checkered CNN (20+ stars) - my deep learning technique, checkered subsampling.
- Passenger Screening Model (20+ stars) - solution to $1.5 million passenger screening challenge.
- GauchoMap Chrome Extension - 200+ users when I was maintaining it (2016).

Education

- 5-year accelerated B.S. + M.S. program @ University of California, Santa Barbara
- Master of Computer Science (2018)
- Bachelor of Computer Science (2017)

Miscellaneous

- Top-30(?) Dance Dance Revolution player in California.
- 2nd-degree black belt in Soo Bahk Do.
- First-generation college graduate.
- GRE scores: 5.5 W, 166 Q, 162 V (90 - 99th percentile)
- Languages: English (native), Farsi (native, but very rusty)